Girls' Education ROI Model - Kenya
Written Analysis

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INTRODUCTION

Access to secondary education for girls in Sub-Saharan Africa remains one of the most studied topics in development economics. Economists have long argued that investments in girls' education generate substantial economic returns, not only for individuals but also for households, communities, and national economies. This analysis applies a discounted cash flow framework to quantify those returns in the context of rural Kenya, connecting academic research on returns to education with financial modeling methods used in development finance and impact investing.

The motivation for this project is personal. Through the Msichana Bora Initiative in Kitui County, Kenya, I conducted field research on barriers to girls' school retention and observed directly how economic constraints drive dropout decisions. I used finance to address a problem I have witnessed firsthand. This model attempts to translate that lived context into a financial argument, one that development finance practitioners, impact investors, and policymakers can engage with on their own terms.

DATA AND ASSUMPTIONS

The model draws on two primary academic sources:

1. Psacharopoulos & Patrinos (2018) provide the foundational wage premium estimate. Their review of 1,120 estimates across 139 countries finds a global average return of approximately 9% per additional year of schooling, with women consistently experiencing above-average returns. This analysis uses 10% as the base-case annual wage premium and 12% as the girls-specific high case.

2. Ozier (2015), in a regression discontinuity study of secondary schooling in Kenya, finds causal evidence that secondary education increases human capital and shifts workers out of low-skill informal employment. This provides Kenya-specific justification for applying the global wage premium to a rural Kenyan context.

Baseline assumptions:
- Annual income without secondary education: KES 70,000 (estimate, rural informal sector)
- Cost per year of secondary school: KES 20,000 (public day school estimate)
- Working life: 35 years
- Discount rate: 8%

Income and cost figures are working estimates. Future iterations of this model should incorporate Kenya National Bureau of Statistics Labour Force Survey data for greater precision.

METHODOLOGY

The model calculates ROI in five steps:

1. Total schooling cost: KES 20,000/yr × 4 years = KES 80,000.

2. Projected earnings with secondary education were estimated by applying the assumed annual wage premium to baseline earnings over four years of secondary schooling.
KES 70,000 × $(1.10)^4$ = KES 102,487/yr

3. Additional annual income: KES 102,487 − KES 70,000 = KES 32,487/yr.

4. Net present value of earnings stream: KES 32,487/yr discounted at 8% over 35 working years = KES 378,622.

5. Discounted ROI: (KES 378,622 − KES 80,000) / KES 80,000 × 100 = 373%.

FINDINGS

Scenario	Wage Premium	NPV of Earnings	Discounted ROI
Low	5%	KES 175,814	120%
Base	10%	KES 378,622	373%
High	12%	KES 467,888	485%

Under the assumptions specified in this model, the base-case scenario produces an estimated discounted ROI of 373%, meaning every KES 1 invested in a girl's secondary education returns KES 4.73 in present value terms over her working life. Even the conservative scenario (5% wage premium) produces a 120% return, confirming that the investment case for girls' education holds across a wide range of assumptions.

The high case (12% girls-specific premium) produces a 485% ROI, consistent with World Bank research showing women experience above-average returns to schooling globally.

These results remained positive across all sensitivity scenarios, suggesting that the investment case for girls' education is robust to reasonable variation in wage premium assumptions.

LIMITATIONS

This model has three important limitations to acknowledge:

1. Baseline income and schooling cost figures are estimates. Precise figures from KNBS Labour Force Survey data would improve accuracy.

2. The model assumes the full wage premium is realized in formal employment. In rural Kenya, where informal employment dominates, actual realized premiums may be lower - closer to the low-case scenario.

3. The model captures only direct earnings gains. It does not account for secondary benefits of girls' education including reduced fertility rates, improved child health outcomes, and lower intergenerational poverty - all of which would increase the true social ROI substantially above the figures presented here.

CONCLUSION

Across all scenarios modeled, girls' secondary education in rural Kenya generated positive discounted returns. While the precise magnitude of these returns depends on the assumptions employed, the analysis suggests that investments in girls' education remain among the most economically compelling opportunities in development finance.

REFERENCES

Psacharopoulos, G., & Patrinos, H. A. (2018). Returns to investment in education: A decennial review of the global literature. Education Economics, 26(5), 445–458.

Ozier, O. (2015). The Impact of Secondary Schooling in Kenya: A Regression Discontinuity Analysis. World Bank Policy Research Working Paper 7384.